

Model Research & Specifications

Use Case: Telecom Customer Churn Prediction

1 Metadata & Project Overview

Use Case Customer Churn Prediction (Telecommunications / ISP)

Target Users Customer Value Management Analysts, Data Scientists

Objective. Generate a realistic synthetic panel dataset simulating customer behaviour over time—including subscription, usage, support interactions, and payments—for churn prediction modelling.

Definition of “Realistic”:

- Panel data with monthly observations
- Heterogeneous usage patterns across plans
- Noisy and incomplete data (missingness, outliers, logging artefacts)
- Latent behavioural drivers (e.g. satisfaction, loyalty)
- Class imbalance in churn ($\approx 15\text{--}25\%$)

2 Data Schema Design (Relational Model)

Tables & Cardinality

Table	PK	FK	Description	Cardinality
customers	customer_id	—	Static demographics & acquisition info	1 row per customer
subscriptions	subscription_id	customer_id	Plan & contract info	1 row per customer
usage_monthly	usage_id	customer_id	Monthly usage (data, calls, SMS)	12 rows / customer / year
support_tickets	ticket_id	customer_id	Customer service interactions	0–5 tickets / customer / month
payments	payment_id	customer_id	Billing and payment info	1 row / customer / month

Relationship Notes:

- Each customer has one subscription.
- Each customer has multiple monthly usage records, payments, and tickets.
- One-to-many relationships: `customers` $\rightarrow$ `usage`, `support_tickets`, `payments`.

3 Data Dictionary (Condensed)

customers

Feature	Type	Description
customer_id	int	Unique ID
age	int	18–80
gender	categorical	Male / Female
region	categorical	Region code
income_bracket	categorical	Low, Mid, High
signup_channel	categorical	Web, App, Store

subscriptions

Feature	Type	Description
subscription_id	int	Unique ID
customer_id	int	FK to customers
plan_type	categorical	Basic, Standard, Premium
monthly_fee	float	Base fee + noise
contract_type	categorical	Prepaid, Postpaid
discount_pct	float	Discount applied

usage_monthly

Feature	Type	Description
usage_id	int	Unique ID
customer_id	int	FK
month	int	1–12
data_usage_gb	float	Monthly GB usage
call_minutes	int	Monthly call minutes
sms_count	int	SMS messages sent
roaming_usage	float	GB used while roaming

support_tickets

Feature	Type	Description
ticket_id	int	Unique ID
customer_id	int	FK
month	int	1–12
ticket_count	int	Number of tickets
issue_type	categorical	Billing, Technical, Service

payments

Feature	Type	Description
payment_id	int	Unique ID
customer_id	int	FK
billing_month	int	1–12
amount_due	float	Monthly fee minus discount
amount_paid	float	Actual payment
delay_days	float	Days delayed

Total features: 25+ across 5 tables.

4 Latent Variables (Omitted Variables) ▷ Internal Only

Variable	Distribution	Description	Affects	Justification
CSAT (Satisfaction)	Beta(2, 2)	Overall satisfaction	churn, tickets, usage	Beta bounded [0,1]; flexible for satisfaction studies (Fader & Hardie, 2007)
Propensity_to_Churn	Beta(1, 3)	Loyalty / switching tendency	churn, tenure	Skewed toward low churn risk (Verbeke et al., 2012)
Payment_Reliability	Beta(3, 1)	On-time payment likelihood	delay_days	Skewed toward reliable payers
Tech_Savviness	Uniform(0, 1)	Digital fluency	usage, tickets	Uniform prior to induce variance in digital behaviour

Note: Latent variables are generated internally and removed before final export.

5 Structural Model (Mathematical Specification)

5.1 Demographics

$$\text{age}_i \sim \mathcal{N}(35, 10^2), \quad \text{age}_i \in [18, 80] \quad (1)$$

$$\text{income}_i \sim \text{Categorical}([0.4, 0.4, 0.2]) \quad (\text{Low, Mid, High}) \quad (2)$$

Justification: Age ~ Normal is common in telecom datasets; income skewed to reflect population (Verbeke et al., 2012).

5.2 Tenure

$$\text{tenure}_i \sim \text{Poisson}(\lambda = 12 + 24 \cdot (1 - \text{propensity}_i)) \quad (3)$$

Reasoning: Poisson is suitable for months of tenure; higher loyalty $\Rightarrow$ longer tenure.

5.3 Subscription Model

$$\text{plan}_i \sim \text{Categorical}([0.5, 0.3, 0.2]) \quad (4)$$

$$\text{monthly_fee}_i = \text{base}(\text{plan}_i) + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, 50) \quad (5)$$

5.4 Time Dimension

$$t = 1, 2, \dots, 12 \quad (\text{months}) \quad (6)$$

Panel dataset capturing evolving customer behaviour.

5.5 Usage Behaviour

$$\mu_{i,t} = 5 + 10 \cdot \text{plan_multiplier} + 5 \cdot \text{tech}_i \quad (7)$$

$$\text{data_usage}_{i,t} \sim \mathcal{N}(\mu_{i,t}, 4^2) \quad (8)$$

$$\text{call_minutes}_{i,t} \sim \text{Poisson}(100 + 50 \cdot (1 - \text{tech}_i)) \quad (9)$$

$$\text{sms}_{i,t} \sim \text{Poisson}(50) \quad (10)$$

Justification: Count variables $\Rightarrow$ Poisson; usage and calls affected by tech-savviness.

5.6 Support Tickets

$$\lambda_{i,t} = 2 + 3(1 - \text{CSAT}_i) + (1 - \text{tech}_i) \quad (11)$$

$$\text{tickets}_{i,t} \sim \text{Poisson}(\lambda_{i,t}) \quad (12)$$

5.7 Payments

$$\text{delay}_{i,t} \sim \text{Exponential}(\lambda = 5 \cdot \text{payment_reliability}_i) \quad (13)$$

$$\text{amount_due}_{i,t} = \text{monthly_fee}_i \cdot (1 - \text{discount}_i) \quad (14)$$

$$\text{amount_paid}_{i,t} = \text{amount_due}_{i,t} - \epsilon, \quad \epsilon \sim \mathcal{N}(0, 20) \quad (15)$$

5.8 Churn Model

$$P(\text{churn}_i = 1) = \sigma \left(\underbrace{-3}_{\text{intercept}} + \underbrace{-0.02}_{\text{tenure}} \cdot \text{tenure}_i + \underbrace{-2}_{\text{CSAT}} \cdot \text{CSAT}_i + \underbrace{1.5}_{\text{propensity}} \cdot \text{propensity}_i + \underbrace{0.3}_{\text{tickets}} \cdot \overline{\text{tickets}_i} \right) \quad (16)$$

$$\text{churn}_i \sim \text{Bernoulli}(p_i) \quad (17)$$

Coefficient Rationale:

- Intercept calibrated to achieve $\approx 20\%$ churn.
- CSAT: strong negative effect; tenure: small negative effect.
- Tickets mildly increase churn.
- Propensity represents latent switching risk.

6 Sampling Workflow

1. Generate customer demographics.
2. Generate latent variables.
3. Assign subscription plans & fees.
4. Compute tenure.
5. For each month $t = 1, \dots, 12$:
 - a. Generate usage (data, calls, SMS).
 - b. Generate support tickets.
 - c. Generate payments (`amount_due`, `amount_paid`, `delay_days`).
6. Aggregate statistics (monthly averages for churn model).
7. Compute churn probability.
8. Sample churn outcome.
9. Inject realism noise & outliers.
10. Drop latent variables for final export.

7 Realism Features

Feature	Implementation	Reasoning
Missing values	5% usage, 3% income	Real-world incomplete data
Outliers	Top 1% usage $\times 3$; call_minutes $\times 2$	Heavy users exist
Noise	Gaussian noise on fees & payments	Measurement error
Rounding	Payments rounded to nearest peso	Realistic accounting
Class imbalance	15–25% churn	Reflects real telecom churn rates
Logging errors	Negative delays clipped	Data entry / system errors
Behavioural variance	Tech_savviness & CSAT induce heterogeneity	Simulates realistic customer segments

8 Behavioural / Business Logic Notes

- **Premium users:** Higher fee $\rightarrow$ more data $\rightarrow$ lower churn (expected behaviour).
- **CSAT:** Drives support interactions $\rightarrow$ realistic negative correlation with churn.
- **Propensity_to_churn:** Models latent loyalty $\rightarrow$ explains variance not captured by observed features.
- **Payment reliability:** Affects late payments $\rightarrow$ correlates with churn indirectly.

References

- [1] Fader, P. S., & Hardie, B. G. S. (2007). How to project customer retention. *Journal of Interactive Marketing*.
- [2] Verbeke, W., Martens, D., & Baesens, B. (2012). Social network analysis for customer churn prediction. *Applied Soft Computing*.
- [3] Hilbe, J. M. (2011). *Negative Binomial Regression*. Cambridge University Press.